LITERATURE SURVEY FOR MULTIMODEL EMOTION RECOGNITION

Reference Paper	Dataset used	Features extracted	Tech Stack	Framework
Multimodal emotion recognition based on hybrid attention networks	**IEMOCAP**: Interactive Emotional Dyadic Motion Capture database **MELD**: Multimodal EmotionLines Dataset **RAVDESS**: Ryerson Audio-Visual Database of Emotional Speech and Song	**Audio Features**: Mel-frequency cepstral coefficients (MFCCs), chroma features, and spectral contrast. **Visual Features**: Facial landmarks and Action Units (AUs) extracted using facial recognition tools. **Text Features**: Word embeddings derived from pre-trained models such as GloVe and BERT.	**Programming Languages**: Python **Libraries**: TensorFlow, Keras, OpenCV, librosa, NLTK/Transformers, Bert	**Supervised Learning:** Classification algorithms like SVM, Random Forest, and Logistic Regression. **Deep Learning:** CNNs for image processing, RNNs, and LSTMs for sequential data, and Transformers for advanced NLP tasks. **Ensemble Methods:** Combining multiple models to improve overall performance.
Facial expression and body gesture emotion recognition: A systematic review on the use of visual data in affective computing.	JAFFE database and CIX database	Matrices and optical flow images were used as inputs for Convolutional Neural Networks (CNNs) to learn high-level features.	ResNet50, WKELM	Support Vector Machines (SVM) and Weber Local Descriptor.
a change of heart: improving speech emotion recognition through speech-to-text modality conversion	MELD (Multimodal EmotionLines Dataset), which consists of audio-visual clips from the TV series "Friends."	**Speech Features Acoustic Features:** Extracted using tools like OpenSMILE or similar libraries, which might include prosodic features (pitch, energy, speaking rate), spectral features (MFCCs, spectral flux), and voice quality features (jitter, shimmer).	**Programming Languages:** Likely Python for most of the implementation. **ASR System:** Vosk API for speech-to-text conversion. **Text Classification:** RoBERTa-base pre-trained model, fine-	**ASR Framework:** Vosk API, which provides an offline speech recognition toolkit. **Text Classification Framework:** Hugging Face's implementation of RoBERTa within either PyTorch or TensorFlow.

		Text Features Transcripts: Text obtained from the Automatic Speech Recognition (ASR) system (Vosk API) or the gold transcripts in the case of perfect ASR. **Text Embeddings:** Generated using a pre-trained RoBERTa-base model fine-tuned on the emotion-labeled transcripts.	tuned for emotion recognition.	
Multi-Label Multimodal Emotion Recognition With Transformer-Based Fusion and Emotion-Level Representation Learning	Interactive Emotional Dyadic Motion Capture (IEMOCAP) and CMU Multimodal Opinion Sentiment and Emotion Intensity (CMU-MOSEI).	Features include word embeddings from text transcription, cropped faces from video frames, Mel spectrogram chunks from audio signals, and emotion-level embeddings.	PyTorch	ALBERT-base model, MTCNN for face detection
Multimodal emotion recognition using cross modal audio-video fusion with attention and deep metric learning	IEMOCAP CMU-MOSI: Carnegie Mellon University Multimodal Opinion Sentiment and Emotion Intensity. MSP-IMPROV: Multimodal corpus of Spontaneous Emotional Speech.	**Text Features:** Word embeddings and other linguistic features extracted from the text modality. **Audio Features:** Acoustic features such as Mel Frequency Cepstral Coefficients (MFCCs) and pitch. **Visual Features:** Facial expression features extracted using convolutional neural networks (CNNs).	**Programming Languages:** Python **Libraries/Frameworks:** PyTorch, NumPy and pandas, scikit-learn, OpenCV and Dlib	Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, for sequential data processing. Convolutional Neural Networks (CNNs) for visual feature extraction. Deep Metric Learning to improve the discriminative power of the learned representations.

Dataset Available:

Dataset	Samples	Language	Type	Recording Environment	Emotions	Data Sources	Speakers
IEMOCAP (2008) [24]	10,039	English	Acted and Natural	In the lab	Discrete: anger, sadness, happiness, disgust, fear, surprise, frustration, excited, and neutral states; Continuous: VAD (activation, valence, and dominance)	Recored	10
Youtube dataset (2011) [25]	47	English	Natural	In the wild	Discrete: positive, negative, and neutral	Youtube	47
MOUD (2013) [26]	498	Spanish	Natural	In the wild	Discrete: positive, negative, and neutral	Youtube	80
ICT-MMMO (2013) [27]	370	English	Natural	In the wild	Discrete: strongly negative, weakly negative, neutral, weakly positive, and strongly positive	Youtube and ExpoTV	370
CMU-MOSI (2016) [28]	2199	English	Natural	In the wild	Discrete: positive, negative	Youtube	89
NNIME (2017) [29]	6701	Chinese	Natural	In the lab	Discrete: anger, sadness, happiness, frustration, neutral, and surprise; Continuous: VA (valence and arousal)	Record	44
CMU-MOSEI (2018) [30]	23,453	English	Natural	In the wild	Discrete: anger, happiness, disgust, sadness, fear, and surprise	Youtube	1000
OMG(2018) [31]	7371	English	Acted and Natural	In the wild	Discrete: anger, happiness, disgust, sadness, fear, surprise, and neutral; Continuous: VA (valence and arousal)	Youtube	N/A
MELD (2019) [32]	13,708	English	Acted	In the wild	Discrete: (1) joy, sadness, anger, fear, disgust, surprise, and neutral; (2) positive, negative, neutral	TV series Friends	407
SEWA (2019) [33]	2562	Chinese, English, German, Greek, Hungar-ian, and Serbian	Natural	In the wild	Continuous: VA (valence and arousal)	Record	398
CH-SIMS (2020) [34]	2281	Chinese	Acted	In the wild	Discrete: negative, weakly negative, neutral, weakly positive, and positive	Movies, TV series	N/A

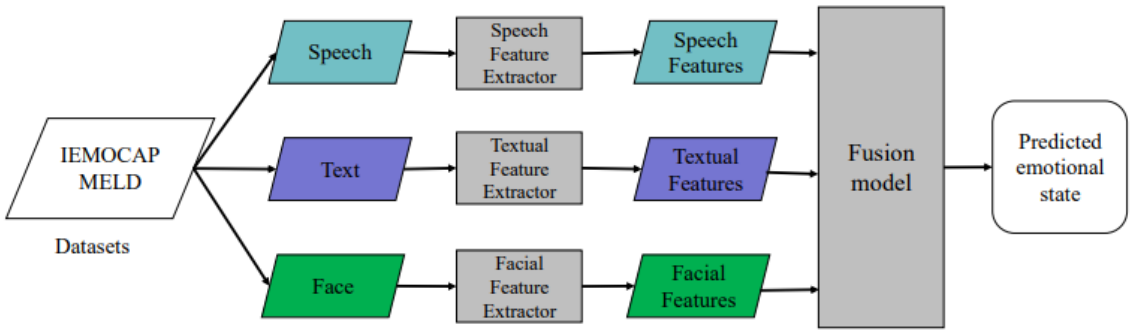

Figure 1. General Framework

Usecases:

Psychological Research: Understanding emotional responses in controlled environments.
Market Research: Analyzing consumer reactions to products or advertisements.
Healthcare: Monitoring emotional states for patient well-being assessment